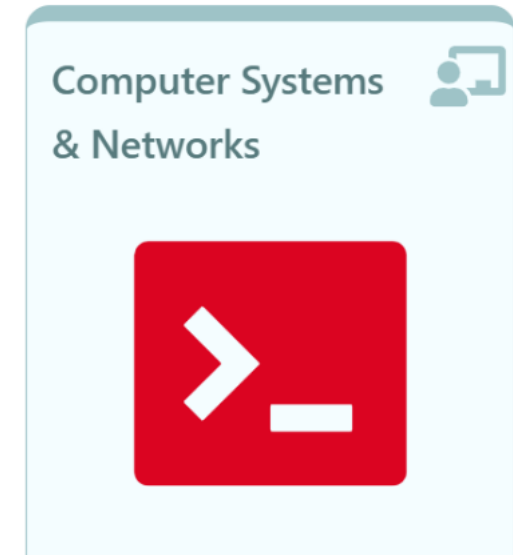
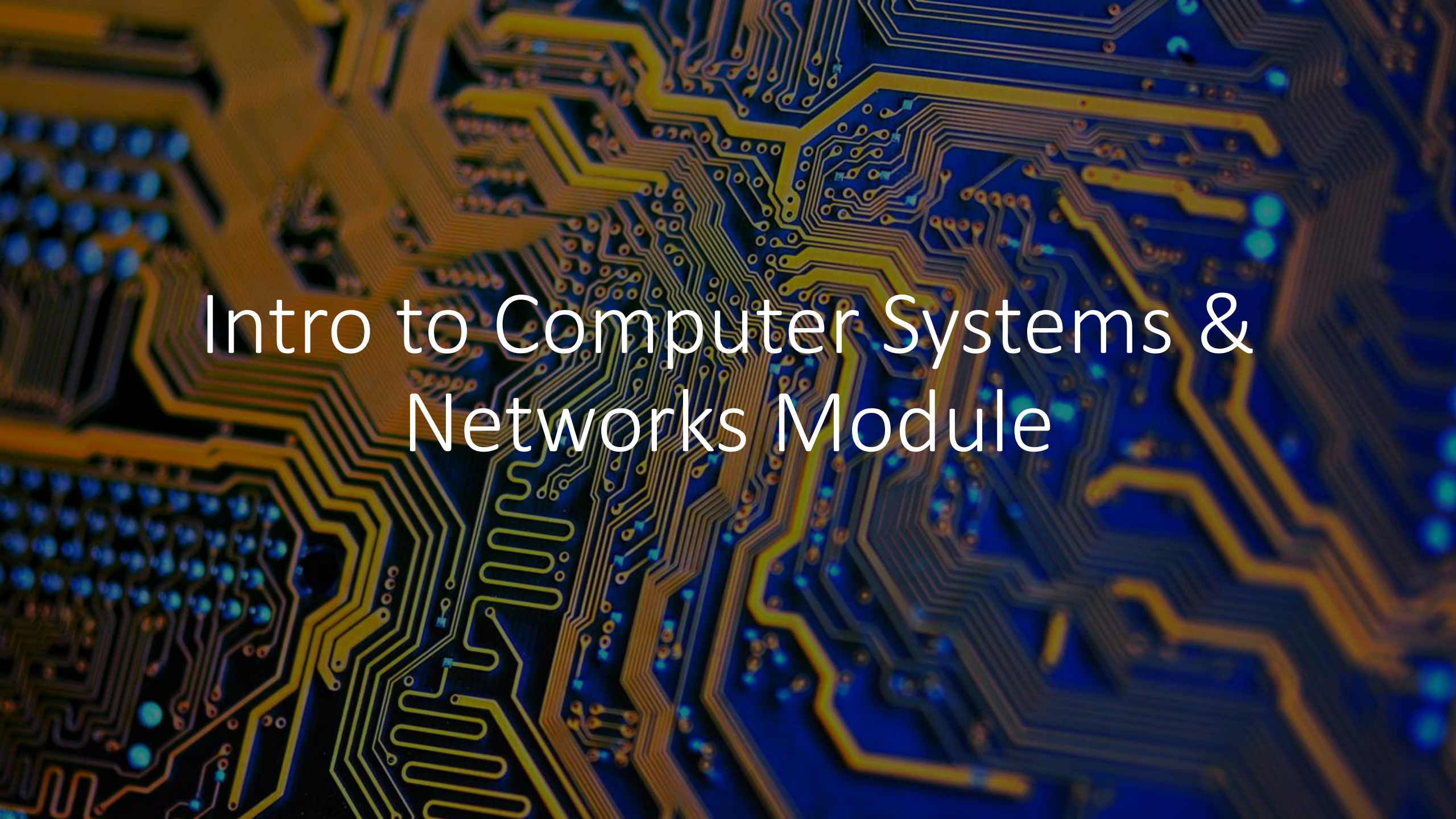


Computer Systems & Networks





Intro to Computer Systems & Networks Module

Contact Me

caroline.cahill@setu.ie

Or

Tag my name on slack
#caroline cahill





Semester 2 2025

Computer Systems & Networks



logic · computer organisation · os · networks · interfaces · sensors

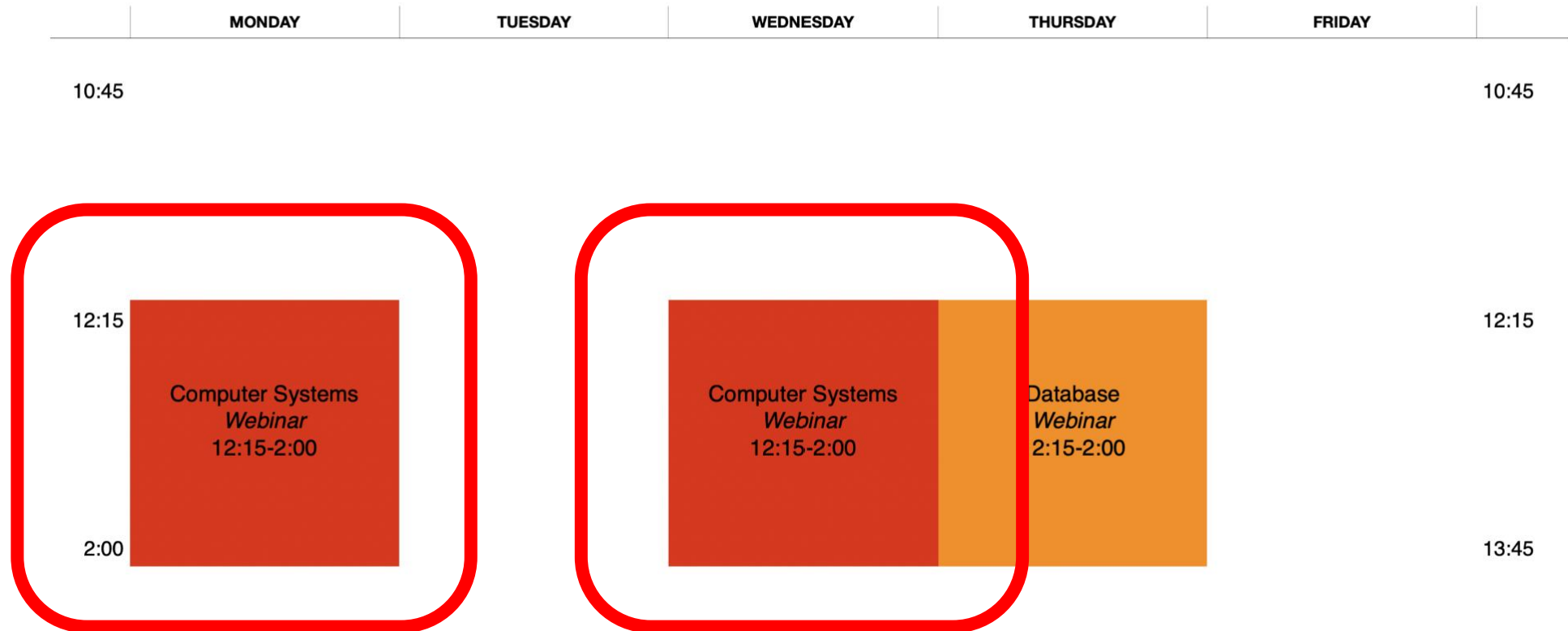
Database



entities · tables · rows · sql · er · nosql

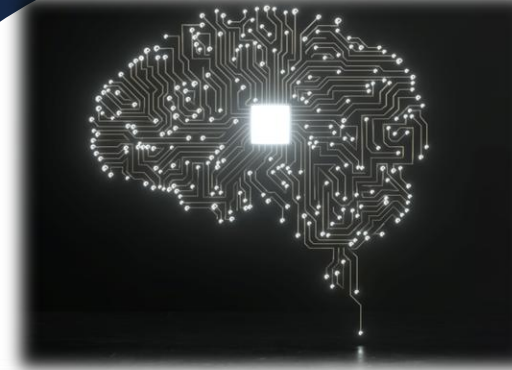
Semester 2		S	M	T	W	T	F	S	Modules
January									
	1	18	19	20	21	22	23	24	comp sys & database
	2	25	26	27	28	29	30	31	comp sys & database
February									
	3	1	2	3	4	5	6	7	comp sys & database
	4	8	9	10	11	12	13	14	
	reading-week	15	16	17	18	19	20	21	comp sys & database
	5	22	23	24	25	26	27	28	comp sys & database
March									
	6	1	2	3	4	5	6	7	comp sys & database
	reading-week	8	9	10	11	12	13	14	comp sys & database
	7	15	16	17	18	19	20	21	
	8	22	23	24	25	26	27	28	comp sys & database
April									
	EASTER	29	30	31	1	2	3	4	comp sys & database
	EASTER	5	6	7	8	9	10	11	
	9	12	13	14	15	16	17	18	comp sys & database
	10	19	20	21	22	23	24	25	comp sys & database
	11	26	27	28	29	30	1	2	comp sys & database
	12	3	4	5	6	7	8	9	comp sys & database
May									
	reading-week	10	11	12	13	14	15	16	
	reading-week	17	18	19	20	21	22	23	

Weekly Webinar Schedule

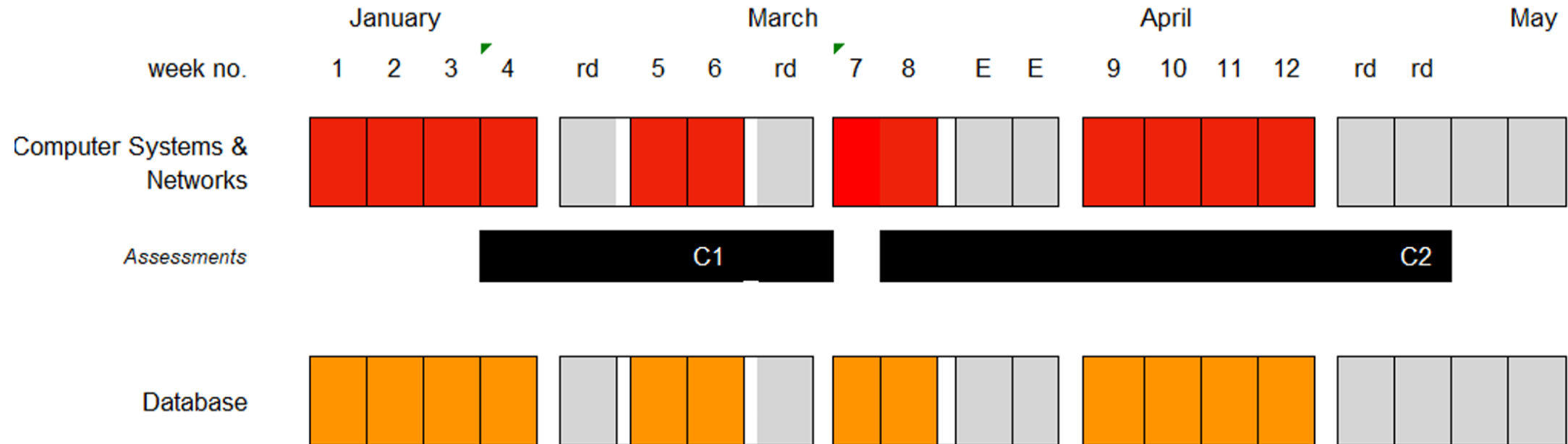


Expectations

caroline.cahill@setu.ie



Semester 2 Assessment Schedule

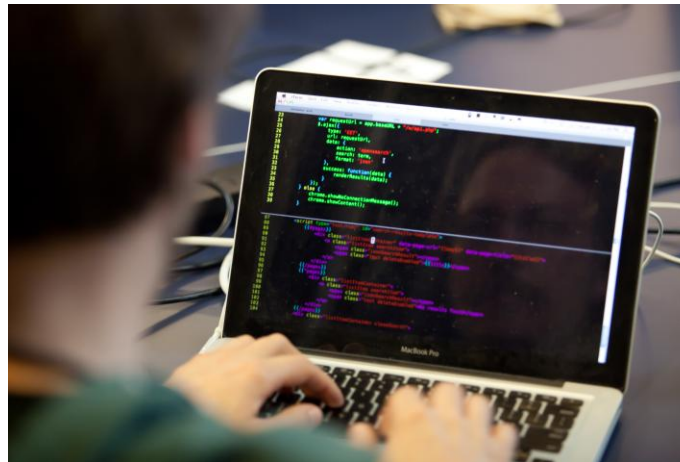


TBCA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams

(try to complete
before your larger
20% CA)



20% Linux Shell scripting

Begins with
5%
Homework



5% InClass Exercises

CA for Caroline's Computer Systems



5% Cisco
NetAcad
chapter exams
(*ongoing*)



Networking Academy

YOU SHOULD HAVE RECEIVED AN EMAIL INVITE FROM ME, SIMILAR TO:



● **Cisco Networking Academy** ☆ **Join Linux Essentials at Waterford Institute of Technolo...** Hello

Module 1 - Introduction to Linux

Chapter 1

Module 2 - Operating Systems

Chapter 2

Chapter 02 Exam

Module 3 - Working in Linux

Chapter 3

Chapter 03 Exam

Module 4 - Open Source Software and Licensing

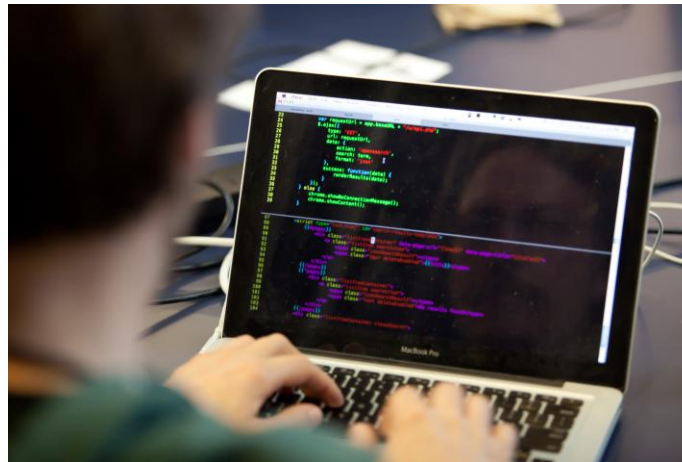
Chapter 4

Chapter 04 Exam

CA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams



20% Linux Shell scripting assessment
(Given *before 1st Reading Wk, Due END 2nd Reading Week*)

Computer Systems & Networks

week no.

123rd456

Assessments

C1

2025

Semester 2

	S	M	T	W	T	F	S	Modules
January								
1	18	19	20	21	22	23	24	comp sys & database
2	25	26	27	28	29	30	31	comp sys & database
February								
3	1	2	3	4	5	6	7	comp sys & database
4	8	9	10	11	12	13	14	
reading-week	15	16	17	18	19	20	21	comp sys & database
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EASTER	29	30	31	1	2	3	4	comp sys & database
EASTER	5	6	7	8	9	10	11	
9	12	13	14	15	16	17	18	comp sys & database
10	19	20	21	22	23	24	25	comp sys & database
11	26	27	28	29	30	1	2	comp sys & database
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reading-week	10	11	12	13	14	15	16	
reading-week	17	18	19	20	21	22	23	
May								

Computer Systems & Networks

logic · computer organisation · os · networks · interfaces · sensors

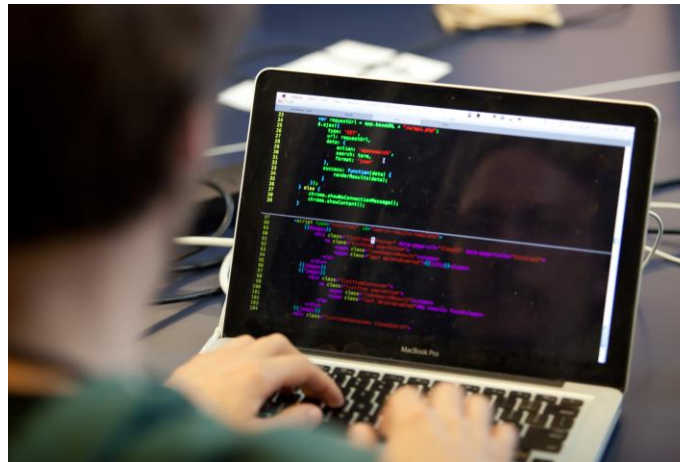
Database

entities · tables · rows · sql · er · nosql

CA for Caroline's Computer Systems



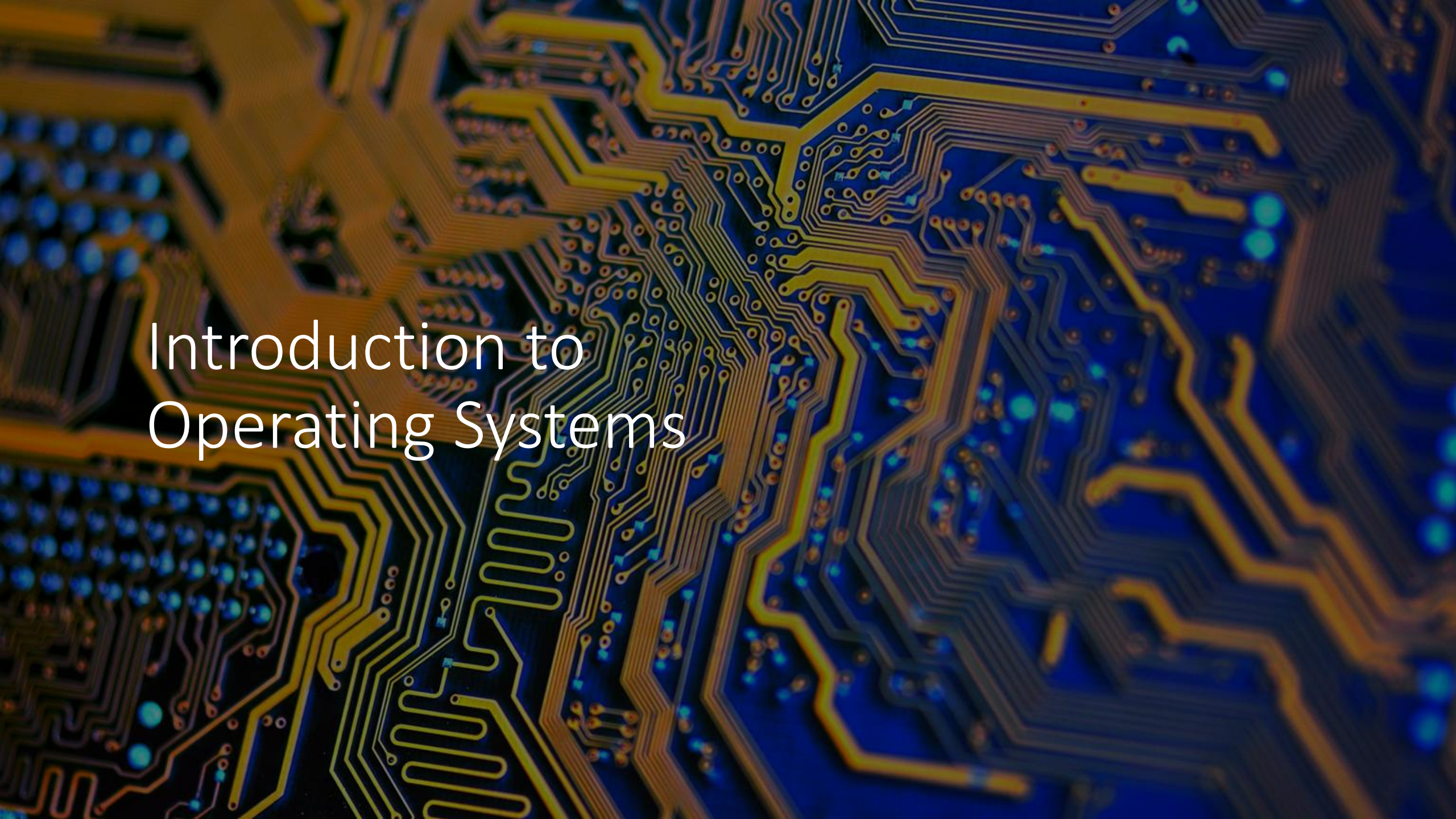
5% Cisco NetAcad
chapter exams



20% Linux Shell scripting

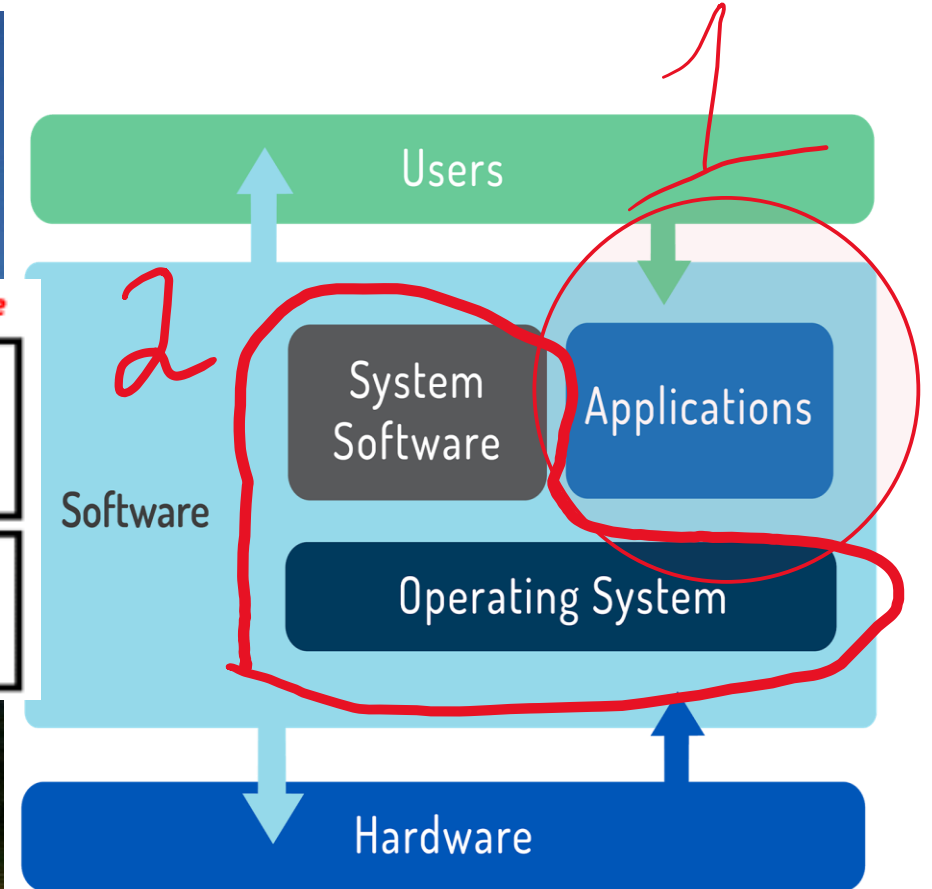
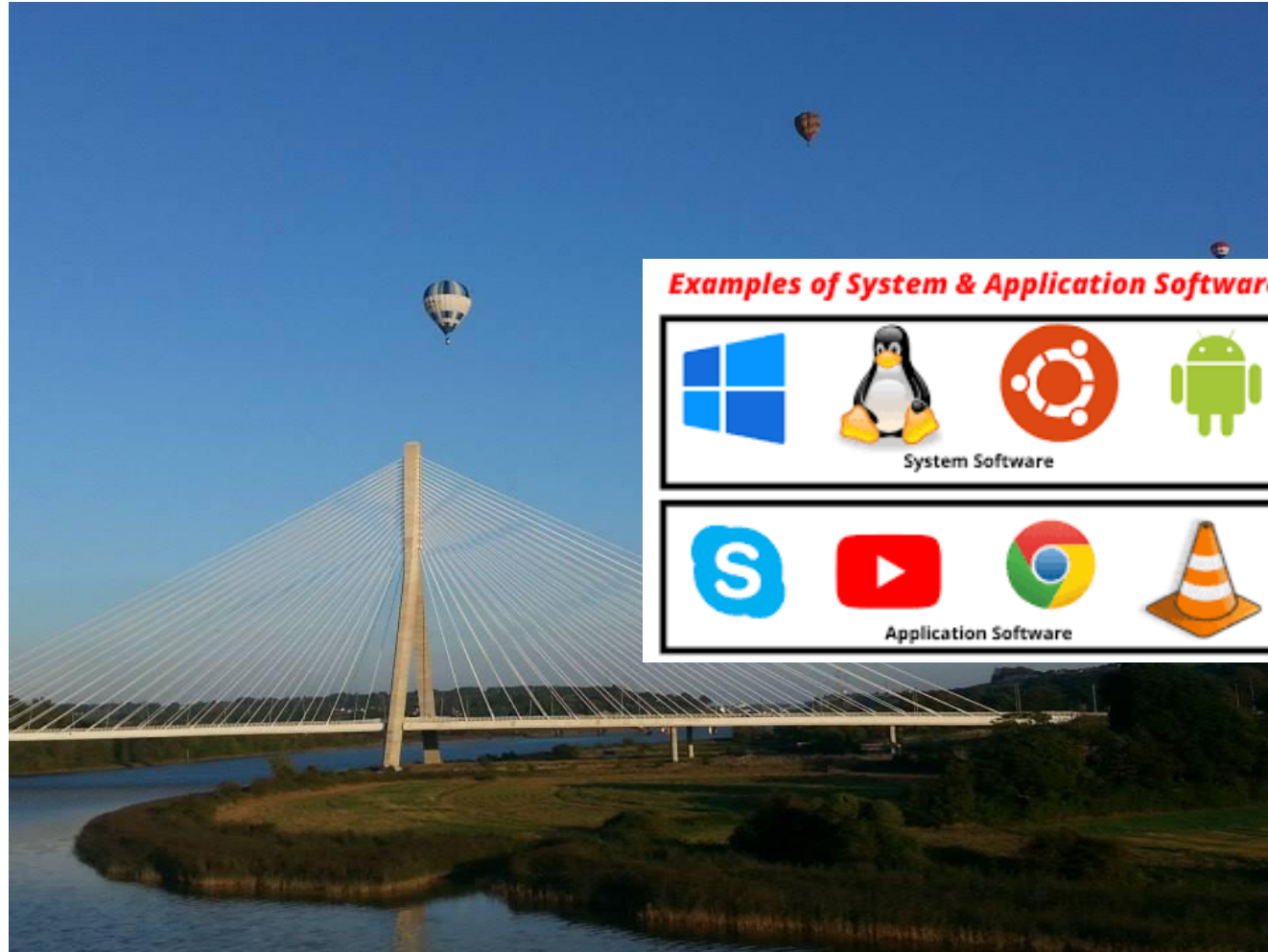


5% Classwork:
completed scripts



Introduction to Operating Systems

Operating System (abbr. OS)



Decision Points when Considering which OS is best

- **Role:** Accessed by 1 user directly (desktop) or many users remotely (server)?
- **Function:** Does it need to run specific software?
- **Life Cycle:** What is the service lifetime?
- **Stability:** Are OS releases *beta* (not tested “in the wild”) or *stable* (tested)?
- **Compatibility:** Is it backwards compatible
- **Cost:** important factor for new systems

Introduction to Linux (as an OS)



Linux is Everywhere!

- Linux is everywhere, Linux jobs are everywhere, there is a shortage of Linux talent, and, Linux skills are in demand in just about every industry and job category.



Default (GPT-3.5)



why is it important to become familiar with shell scripting?



Becoming familiar with shell scripting offers several important benefits, especially for individuals working in IT, software development, data analysis, and system administration. Here are some reasons why it's important to learn shell scripting:

List Of Companies Using Linux And Its Market Share (techdatapark, Oct 2024)

Company	Website	Country	Revenue	Company Size
TechGlobal Corp.	google.com	United States	\$280B	156,500+
E-commerce Solutions Inc.	amazon.com	United States	\$514B	1,500,000+
Digital Marketing Group	meta.com	United States	\$117B	86,000+
Global Education Networks	ibm.com	United States	\$60B	280,000+
Creative Agency Co.	redhat.com	United States	\$1B	18,000+
Tencent	tencent.com	United States	\$81B	112,000+
Dell Technologies	dell.com	United States	\$102B	165,000+
Cisco Systems	cisco.com	United States	\$53B	79,500+
Oracle	oracle.com	United States	\$48B	135,000+

About 90% of Hollywood special effects rely on Linux

Linux is perfect for Broke Students as it's:

- Free
- Runs great on old laptops
- No license keys
- No "your trial has expired" messages

85% of smartphones are Linux-powered

Android is built on the Linux kernel.

NASA uses Linux on the International Space Station.

Android is built on the Linux kernel.

Supercomputers = Linux Monopoly
100% of the world's top 500 supercomputers run Linux.

- Used for climate modeling, nuclear simulations, AI research, and physics experiments.

Most of the Internet Runs on Linux

- ~90% of the world's servers run Linux
- websites like Google, Facebook, YouTube, Netflix, Wikipedia, Reddit

Linux is the **top choice for developers**, especially in:

- Web development
- AI & machine learning
- Cybersecurity
- Cloud computing

Linux is Open Source

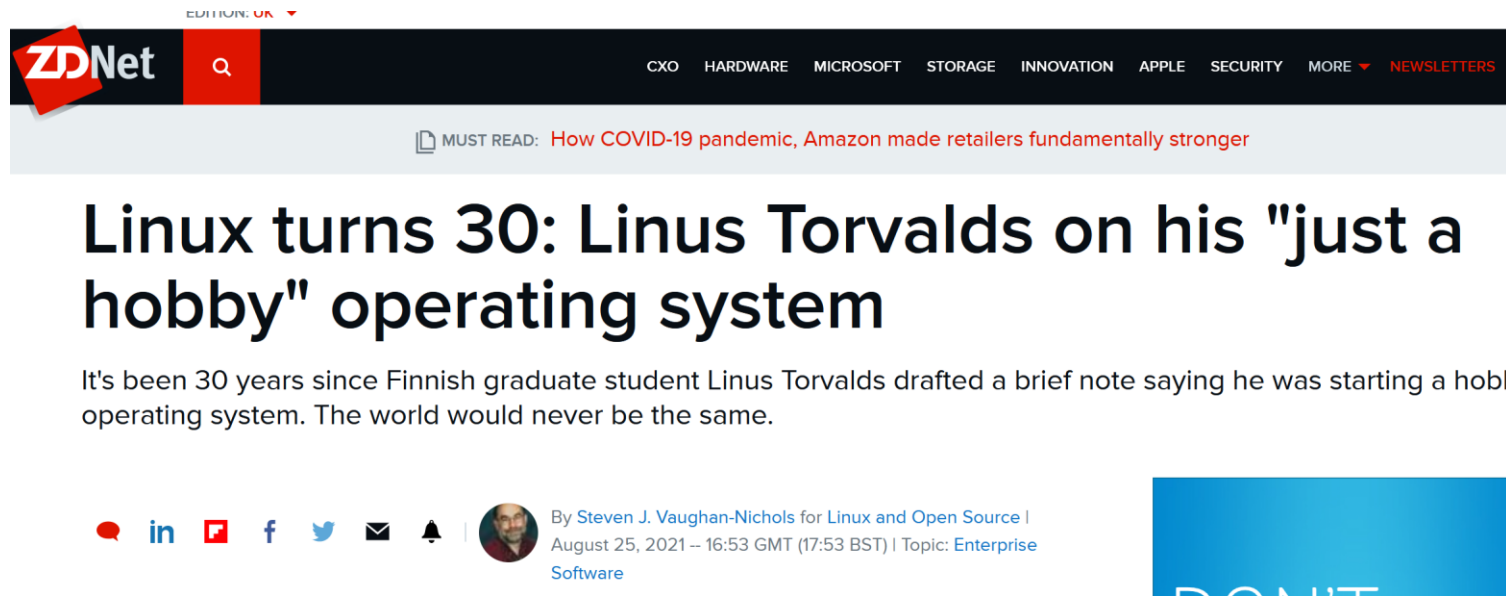
- The development of Linux closely parallels the rise of *open source software*.
- The open source philosophy is that

You have a right to obtain the software **source code** and

You have a right to **modify** it for your own use.

Linux is around a long time

- Linux started in 1991 as a hobby project by Linus Torvalds, a Finnish born computer scientist studying at the University of Helsinki.



The screenshot shows the ZDNet website interface. At the top, there's a navigation bar with the ZDNet logo, a search icon, and links for CXO, HARDWARE, MICROSOFT, STORAGE, INNOVATION, APPLE, SECURITY, MORE, and NEWSLETTERS. Below the navigation bar, a banner reads "MUST READ: How COVID-19 pandemic, Amazon made retailers fundamentally stronger". The main headline is "Linux turns 30: Linus Torvalds on his 'just a hobby' operating system". Below the headline, a sub-headline states: "It's been 30 years since Finnish graduate student Linus Torvalds drafted a brief note saying he was starting a hobby operating system. The world would never be the same." At the bottom of the article preview, there are social media sharing icons (Reddit, LinkedIn, YouTube, Facebook, Twitter, Email, and a bell icon), a small profile picture of Steven J. Vaughan-Nichols, and text indicating the author, date, time, and topic. To the right of the article preview is a blue rectangular placeholder with the word "DON'T" partially visible.

EDITION: UK

ZDNet

CXO HARDWARE MICROSOFT STORAGE INNOVATION APPLE SECURITY MORE NEWSLETTERS

MUST READ: How COVID-19 pandemic, Amazon made retailers fundamentally stronger

Linux turns 30: Linus Torvalds on his "just a hobby" operating system

It's been 30 years since Finnish graduate student Linus Torvalds drafted a brief note saying he was starting a hobby operating system. The world would never be the same.

By Steven J. Vaughan-Nichols for Linux and Open Source | August 25, 2021 – 16:53 GMT (17:53 BST) | Topic: Enterprise Software

DON'T

Linux is a Kernel

- Linux means the *kernel* of the system, which is the central controller of everything that happens on the computer.
- Linux is a combination of software called **GNU/Linux**, which defines the *operating system*.
 - **GNU** is the *free software* that provides open source equivalents of many common UNIX commands.
 - The **Linux** part of this combination is the *Linux kernel*, which is the core of the operating system.

Linux Embraces the CLI

Two basic types of interfaces available that allow you to interact with the operating system



Linux Distros

A distribution refers to the Linux kernel, tools, and suite of applications that come bundled together.

Linux distros refer to an OS composed of open-source programs designed by numerous developers and programmers.



Lubuntu = Lightweight Ubuntu

Ubuntu is one of the popular Linux distributions available for desktops and laptops

- A new version of Ubuntu is released every April and October.

Why Ubuntu?

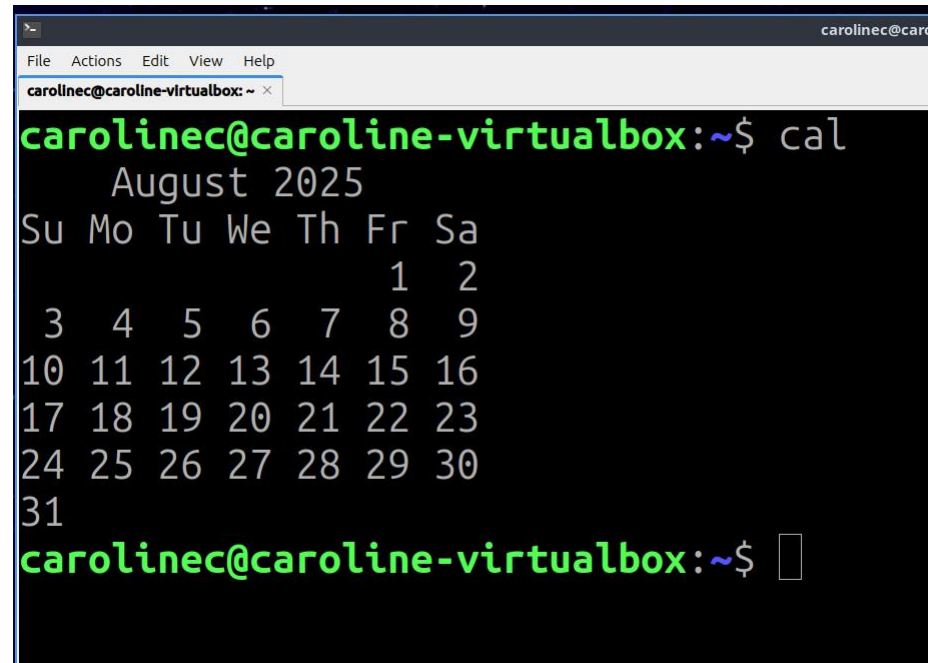
- User-friendly
- There's a thriving Ubuntu community online
- Strong software and hardware support

~~Linux~~ YOU Embrace the CLI

- Enter the Command

\$cal

- to view the built in calendar.



```
carolinec@caroline-virtualbox: ~$ cal
      August 2025
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
carolinec@caroline-virtualbox: ~$
```

```
compsys@compsys-virtualbox:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
```

\$sudo apt install <<program>>

```
carolinec@caroline-virtualbox:~$ cal Jan 2026
    January 2026
Su Mo Tu We Th Fr Sa
      1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

carolinec@caroline-virtualbox:~$
```

```
$sudo apt update
```

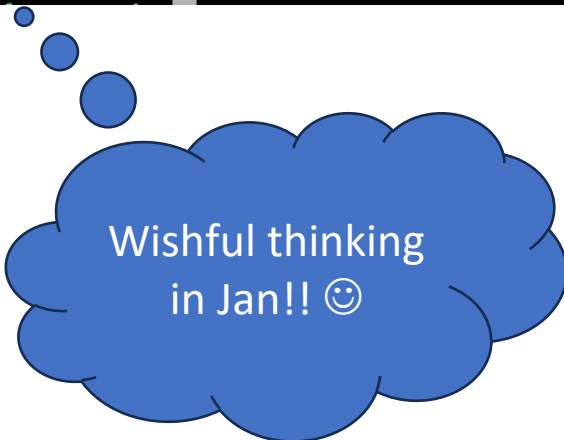
Use this command to update the package database any time

~~Linux~~ YOU Embrace the CLI

- Try look up the current weather in Kilkenny using

\$ansiweather -l *your_town_name*

```
compsys@compsys-virtualbox:~$ ansiweather -l kilkenny  
Weather in Kilkenny: 18 °C - UVI: 5.32 - Wind: 4.02 m/s WNW - Humidity: 75% - Pressure: 1016 hPa
```



Wishful thinking
in Jan!! 😊

Be familiar with the help **manual** available

What does the `-l` stand for in the command line instruction

`$ansiweather -l kilkenny`



~~Linux~~ YOU Embrace the CLI

- Uninstall ansiweather

\$sudo apt-get remove *program-name*



~~Linux~~ YOU Embrace the CLI

- Try out: `$cowsay Well hello there January!`
- *Do you need to install it?*
- Once finished, **uninstall**
- Don't like cows?? `$cowsay -l`
`$cowsay -f turkey Almost 100 days to Christmas, gobble!`
- *The `-l` and `-f` are examples of optional arguments that many CLI commands accept*

~~Linux~~ YOU Embrace the CLI

- Try out:

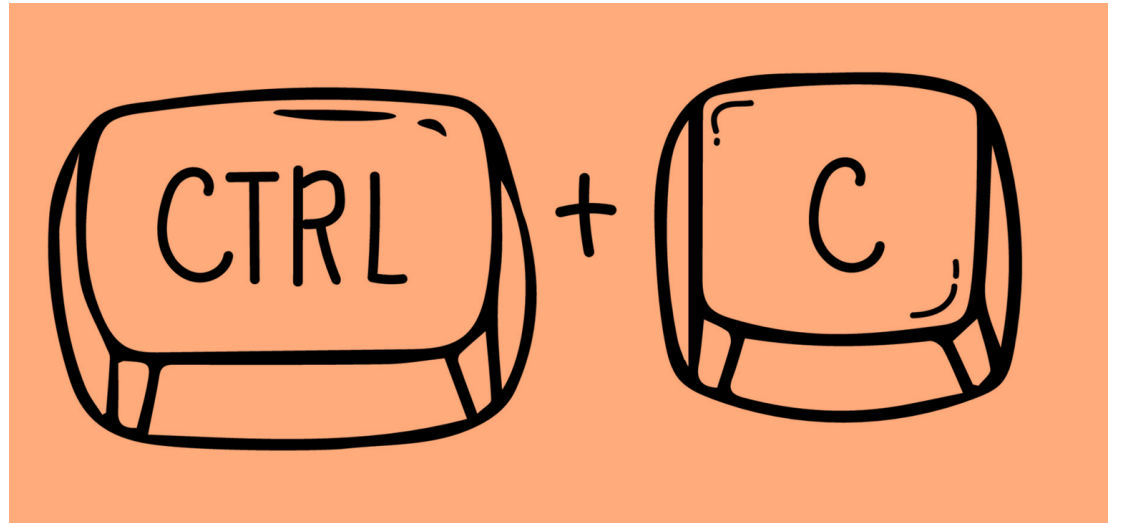
\$cmatrix

Now

\$figlet Lets go

And finally, what happens here
Why?

\$figlet Let's go



Try out today's lab

01: Computer
Systems



Operating System
Linux



NetAcad · Linux as an OS

02: File
Permissions



-	rwX	r--	r--
↓	↓	↓	↓
"-" indicates a file "d" indicates directory "l" indicates a link	Read, write, and execute permissions for the owner of the file	Read, write, and execute permissions for members of the group owning the file	Read, write, and execute permissions for other users

Numbering Systems ·
Permissions · Octal

Lab

```
#!/bin/bash
```

Fundamentals · Command
· Scripts · Permissions